

Multimodal AI Security: Comprehensive Threat Landscape & Defense Architecture Research

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1. Executive Summary

AEGIS as built today secures text-based LLM interactions through seven adaptive defense layers. However, the rapid convergence toward multimodal AI systems — models that process images, audio, video, documents, and tool calls alongside text — has created attack surfaces that text-only defenses cannot address. This research document catalogs the complete multimodal threat landscape across five modalities, surveys the current state of attack research and known defenses, identifies architectural requirements for extending AEGIS, and proposes a future-proof design that anticipates attacker evolution rather than reacting to it.

The core finding is that multimodal attacks exploit a fundamental architectural weakness: content from non-text modalities bypasses text-based security scanners entirely, while the model processes all modalities in a shared semantic space where cross-modal influence is possible. An image containing embedded text instructions, an audio clip with imperceptible adversarial perturbations, or a video with injected frames can all manipulate model behavior without triggering any existing text-based detection.

The immune system architecture does not need to change. The scanner module pattern (L2 innate, L3 adaptive) extends naturally to new modalities. What changes is the set of scanner modules, the embedding models used for the threat vault, and the preprocessing pipeline that normalizes inputs before they reach the model.

2. Modality-Specific Threat Analysis

2.1 Vision / Image Inputs

Vision is the second most widely deployed modality after text, and the most extensively studied attack surface for multimodal models.

2.1.1 Typographic / OCR-Based Prompt Injection

The simplest and most reliable image-based attack: embed text instructions directly into an image that the VLM reads via its inherent OCR capability. Research from October 2025 (Zhu et al., arXiv:2510.09849) demonstrates a systematic text prompt injection algorithm against VLMs that achieves high success rates with minimal computational resources — significantly outperforming gradient-based attacks on large models like LLaVA-Next-72B.

Attack variants include high-contrast visible text (most effective but detectable), low-contrast text matching background color (harder to detect visually but still readable by VLMs), small font text in image margins, text overlaid on busy visual regions, and white-on-white or near-invisible text using CSS-equivalent pixel manipulation.

A Nature Communications study (Clusmann et al., February 2025) tested 594 prompt injection attacks against GPT-4o, Claude-3, Claude-3.5, and Reka Core in oncology applications. Sub-visual prompts embedded in medical imaging data caused models to provide harmful output across all models tested. Low-contrast settings reduced success rates for Claude models (69% to 14% for Claude-3) but remained effective against GPT-4o and Reka Core.

Defense implications for AEGIS: OCR preprocessing of all image inputs before they reach the model, with the extracted text fed through existing L2/L3 text scanners. This is the highest-priority image defense because it reuses the existing detection pipeline.

2.1.2 Steganographic Prompt Embedding

More sophisticated than typographic injection. Malicious instructions are invisibly embedded within images using steganographic techniques — spatial domain (LSB manipulation), frequency domain (DCT coefficient modification), and neural steganography (encoder-decoder networks).

The first comprehensive study (Pathade et al., arXiv:2507.22304, July 2025) achieved an overall attack success rate of 24.3% ($\pm 3.2\%$, 95% CI) across GPT-4V, Claude, and LLaVA, with neural steganography reaching 31.8% while maintaining visual imperceptibility (PSNR > 38 dB, SSIM > 0.94). Tested across 12 datasets and 8 models.

While success rates are lower than typographic injection, steganographic attacks are far harder to detect because they are invisible to human observers and survive typical image processing pipelines (resizing, compression, normalization).

Defense implications for AEGIS: Steganalysis preprocessing — statistical analysis of image pixel distributions to detect anomalous patterns. Image sanitization via recompression/re-encoding that destroys steganographic payloads while preserving visual content. Comparison of model behavior on original vs. sanitized versions of the same image.

2.1.3 Adversarial Perturbation Attacks

Imperceptible pixel-level modifications that cause models to misclassify, generate incorrect outputs, or bypass safety filters. Unlike steganographic attacks (which embed readable instructions), adversarial perturbations exploit the mathematical properties of neural network decision boundaries.

Chain of Attack (CoA), published at CVPR 2025 (Xie et al.), achieves 98.4% targeted attack success rate through step-by-step semantic modifications guided by Targeted Contrastive Matching. The technique produces natural-looking adversarial images that evade detection while precisely manipulating model outputs.

The Multimodal Feature Heterogeneous Attack (MFHA, Nature Scientific Reports, March 2025) targets medical VLMs through triplet contrastive learning, achieving 16.05% average transferability improvement —

demonstrating that safety-critical healthcare AI remains vulnerable.

A critical finding: cross-modal transferability means attacks on the image modality can propagate and degrade predictions in the text modality. This is unique to multimodal models and does not exist in unimodal systems.

Defense implications for AEGIS: Input purification/preprocessing (denoising, smoothing, filtering) before images reach the vision encoder. Adversarial training using augmented image-text pairs. Multi-model consensus (run the same image through multiple vision encoders and flag disagreement).

2.1.4 Cross-Modal Jailbreaking

Attacks that combine visual and textual perturbations to achieve jailbreaks that neither modality alone could accomplish. The "JPS" framework (Jailbreak with Collaborative Visual Perturbation and Textual Steering) demonstrates that coordinated cross-modal attacks are significantly more effective than unimodal attacks.

The CSA (Cloud Security Alliance, May 2025) reported that Mistral's Pixtral models are 60x more prone to generating CSEM-related content when prompted with image-embedded injections compared to GPT-4o and Claude 3.7 Sonnet, and 18-40x more likely to produce dangerous CBRN information under adversarial image inputs. Critically, these risks were triggered by prompt injections within image files, not malicious text inputs — meaning text-only safety filters were completely bypassed.

Defense implications for AEGIS: Cross-modal consistency checking — verify that image content semantics align with text prompt semantics. Flag requests where the image appears benign but the combined image+text input produces anomalous model behavior.

2.2 Audio Inputs

Audio-capable models (GPT-4o voice, Gemini, Qwen2.5-Omni) introduce attack surfaces that are fundamentally different from text and images.

2.2.1 Adversarial Audio Perturbations

Imperceptible modifications to audio waveforms that cause models to transcribe or interpret audio incorrectly. Carlini et al. demonstrated 100% attack success rate on DeepSpeech ASR, producing adversarial examples that transcribe as any target phrase while sounding like normal audio or music to human listeners. Speech can literally be embedded in music that plays normally to humans but is interpreted as commands by AI.

Research on commercial black-box ASR systems (ACM TOPS, 2025) achieved 96% and 80% success rates against Aliyun and Tencentyun ASR APIs respectively using transferable adversarial examples. These adversarial examples triggered actual attack actions on Apple Siri and Xiaomi voice assistants while being inaudible to humans.

Universal Acoustic Adversarial Attacks (EMNLP 2025 Findings) demonstrate attacks that work across multiple tasks (transcription, translation, gender detection) and languages, with a single adversarial audio segment controlling model behavior across different input types.

2.2.2 WhisperInject: Benign Audio, Harmful Output

WhisperInject (arXiv:2508.03365, August 2025) reveals a critical vulnerability: benign-sounding audio can covertly induce harmful text generation. The two-stage framework first discovers model-native harmful responses via RL-PGD, then embeds them into imperceptibly perturbed audio. Achieves over 86% success across state-of-the-art audio-language models including Qwen2.5-Omni (3B and 7B) and Phi-4-Multimodal — even models that underwent extensive RLHF safety tuning.

This represents a new class of threat: the audio sounds completely normal, but the model generates harmful content. Text-based output filters would need to catch the harmful generation after the fact, but by then the audio has already influenced the model's reasoning.

2.2.3 Voice Cloning and Authentication Attacks

Voice cloning from as little as 3 seconds of recorded speech can bypass voice authentication systems. Voice phishing attacks surged 442% in 2025. E2E-VGuard (NeurIPS 2025) addresses this by using adversarial perturbations defensively — adding imperceptible noise to voice recordings that disrupts synthesis while preserving human intelligibility.

Defense implications for AEGIS (audio):

- Audio-to-text transcription as preprocessing step, then run transcript through existing text scanners (analogous to OCR for images)
- WaveGuard-style defense: process audio through transformation function, compare transcriptions of original vs. transformed audio — flag as adversarial if transcriptions diverge beyond threshold
- Spectral analysis to detect adversarial perturbations by comparing audio spectral characteristics against known benign distributions
- Liveness detection for voice authentication use cases
- Audio sanitization via re-encoding that destroys adversarial perturbations

2.3 Video Inputs

Video-capable models (Gemini 1.5/2.5 Pro, GPT-o4) process temporal sequences of frames, creating attack surfaces that combine image-based attacks with temporal dynamics.

2.3.1 Frame-Level Injection

Adversarial frames or text-bearing frames injected at specific positions in a video stream. A medRxiv study (July 2025) evaluated Gemini 1.5 Pro, Gemini 2.5 Pro, GPT-o4-mini-high, and Qwen 2.5-VL across eleven surgical decision support tasks, demonstrating susceptibility to both textual and visual prompt injection attacks embedded in surgical video.

Because video models typically sample frames rather than processing every frame, attackers can target the sampling pattern — injecting adversarial content precisely at frames likely to be selected.

2.3.2 Temporal Consistency Attacks

Attacks that exploit the temporal dimension: gradually shifting content across frames to condition model behavior (similar to multi-turn text escalation, but in the visual domain), embedding information across multiple frames that only becomes meaningful when aggregated, and interleaving benign and adversarial frames to evade per-frame detection.

Defense implications for AEGIS (video):

- Frame extraction and per-frame image scanning using image defense pipeline
- Temporal consistency analysis — flag videos where frame-to-frame semantic content shifts anomalously
- Text extraction from video frames (OCR on sampled frames)
- Frame sampling randomization to defeat targeted frame injection

2.4 Document / File Inputs

PDFs, Word documents, spreadsheets, and other structured files processed by multimodal models.

2.4.1 Hidden Content Injection

Documents can contain hidden text (white-on-white, zero-font-size, metadata fields, comments, tracked changes, hidden layers in PDFs) that is invisible to human readers but processed by models. This is directly flagged in the Stack document: "adversaries can embed content invisible to human visitors (hidden text, white-on-white CSS, invisible divs, metadata fields, image alt text, schema markup) that agents ingest as legitimate data."

2.4.2 Polyglot Files

Files that are valid in multiple formats simultaneously (e.g., a file that is both a valid PDF and a valid ZIP containing executable code). When processed by models that handle multiple file types, polyglot files can trigger unexpected parsing behavior.

Defense implications for AEGIS (documents):

- Document sanitization: strip metadata, comments, hidden text, zero-width characters before model processing
- Render-then-scan: convert documents to images, OCR the rendered output, compare against raw extracted text — hidden content shows up in raw extraction but not in rendered output
- File format validation: reject polyglot files and files with mismatched magic bytes vs. extension

2.5 Tool Use / Function Calling

Not a modality per se, but an interaction pattern that creates distinct attack surfaces.

2.5.1 Jailbreaking via Function Calling

Research published at COLING 2025 demonstrates that function calling creates a distinct jailbreak vector. By injecting adversarial function definitions (e.g., a function named "WriteNovel" with a description containing harmful instructions), attackers trigger the model to call the jailbreak function without any additional jailbreak methods in the prompt. Safety alignment in chat mode does not transfer to function calling mode.

2.5.2 Tool Poisoning and Puppet Attacks

ToolCommander (NAACL 2025) demonstrates manipulation of tool-calling systems by injecting adversarial tools that, when retrieved and invoked, exploit tool responses to manipulate the scheduling process — enabling privacy theft, denial-of-service, and unscheduled tool-calling attacks.

AgentBound (arXiv:2510.21236) identifies multiple MCP server attack categories: tool poisoning (manipulating tool descriptions to compel malicious actions), puppet attacks (manipulating execution semantics of legitimate tool calls), and rug pull attacks (initially benign servers that later modify descriptions).

Agent Security Bench (ICLR 2025) benchmarks 27 attack/defense combinations across 13 LLM backbones, finding highest average attack success rates exceeding 84.30%, while existing defenses are often ineffective.

2.5.3 Cross-Tool Chain Attacks

Attacks that chain multiple tool calls to achieve objectives that no single call would accomplish. For example: use a benign search tool to retrieve content containing injection payloads, then feed that content to a code execution tool.

Defense implications for AEGIS (tool use):

- Function definition scanning: analyze tool descriptions for injection patterns before they enter model context
 - Tool call validation: verify that requested tool calls match the user's authorized capabilities and the conversation's semantic context
 - Output sanitization between tool calls: scan tool outputs before they're fed back to the model
 - Behavioral baselining of tool call patterns: flag unusual sequences, unauthorized tools, capability escalation attempts
 - MCP server integrity verification: validate tool descriptions against known-good manifests
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3. Cross-Modal Attack Patterns

Beyond modality-specific attacks, several patterns exploit the interaction between modalities.

3.1 Modality Laundering

Encoding malicious instructions in one modality to bypass security filters designed for another. Example: text-based injection detection is mature, so attackers encode the injection as text within an image, leveraging the

model's OCR capability to "read" the injection from a modality where no text scanner operates.

3.2 Semantic Anchoring

Using one modality to establish context that makes the model more susceptible to attacks in another modality. Example: an image showing an "official system notification" visual context, combined with a text prompt that references the notification, creates a compound social engineering attack.

3.3 Modality Confusion

Exploiting the model's multimodal fusion layer, where representations from different modalities are combined. Adversarial inputs in one modality can shift the model's internal representations in ways that affect processing of the other modality.

3.4 Progressive Cross-Modal Escalation

Multi-turn attacks that alternate between modalities. Turn 1: benign text establishing rapport. Turn 2: image containing subtle boundary-testing content. Turn 3: text referencing the image to normalize the boundary violation. Turn 4: audio or image containing the actual payload. This is the multimodal equivalent of AEGIS's existing multi-turn sequence analysis, but across modalities.

4. Defense Architecture: Extending AEGIS

4.1 Design Principles for Multimodal Extension

Modality-Agnostic Core, Modality-Specific Scanners: The policy engine (L6), circuit breaker (L7), threat vault (L4), and event bus remain unchanged. New scanner modules plug into L2 (innate) and L3 (adaptive) following the same interface contract: `scan(input) → ScanResult`.

Pre-Model Normalization: Every non-text modality must be normalized to extract any text-equivalent content before the input reaches the model. This is the highest-ROI defense because it reuses the entire existing text detection pipeline.

Defense in Depth Per Modality: Each modality gets its own fast-path (innate) and slow-path (adaptive) scanners, mirroring the dual-path architecture for text.

Cross-Modal Correlation: A new analysis module that examines the relationships between modalities within a single request, detecting semantic inconsistencies that indicate cross-modal attacks.

4.2 Image Security Module

L2 Innate (Fast Path, <10ms):

- OCR text extraction → feed through existing regex engine and blocklist
- Image metadata stripping (EXIF, XMP, IPTC)

- File format validation (magic bytes vs. declared type)
- Image hash comparison against known adversarial image database
- Basic steganalysis: statistical tests on pixel distributions (chi-square, RS analysis)

L3 Adaptive (Slow Path, 20-100ms):

- Vision-language safety classifier (e.g., LLaVA-Guard or similar)
- Deep steganalysis using trained neural network detectors
- Adversarial perturbation detection via input transformation comparison
- Cross-modal consistency check: does the image content align with the text prompt semantics?
- Image sanitization: re-encode image to destroy potential steganographic payloads, compare model behavior on original vs. sanitized

L4 Threat Vault Extension:

- CLIP or SigLIP embeddings of known adversarial images stored alongside text embeddings in FAISS
- Image perceptual hashing (pHash, dHash) for fast similarity matching against known attack images

4.3 Audio Security Module

L2 Innate (Fast Path, <10ms):

- Audio-to-text transcription → feed through existing text scanners
- Audio metadata stripping
- Format validation
- Duration and frequency range checks (detect ultrasonic/infrasonic content outside human hearing)

L3 Adaptive (Slow Path, 50-200ms):

- WaveGuard-style defense: apply audio transformation, compare transcriptions, flag divergence
- Spectral analysis for adversarial perturbation detection
- Voice liveness detection (breathing patterns, micro-pauses, background acoustics)
- Speaker verification against registered voice profiles (for authenticated use cases)
- Audio sanitization via re-encoding through lossy codec (destroys adversarial perturbations)

L4 Threat Vault Extension:

- Audio embeddings (Whisper encoder features or similar) for similarity matching
- Spectral fingerprints of known adversarial audio patterns

4.4 Video Security Module

L2 Innate (Fast Path, <20ms per sampled frame):

- Frame extraction at randomized intervals (defeat targeted frame injection)
- Per-frame OCR → text scanner pipeline
- Per-frame metadata check

L3 Adaptive (Slow Path, 100-500ms):

- Per-frame image safety classification on sampled frames
- Temporal consistency analysis: flag anomalous frame-to-frame semantic shifts
- Audio track extraction → audio security pipeline
- Hidden frame detection (frames with anomalous visual properties vs. neighbors)

4.5 Document Security Module

L2 Innate (Fast Path, <15ms):

- Hidden text extraction (zero-width characters, white-on-white, metadata, comments)
- Comparison of raw text extraction vs. rendered visual output
- File format validation, polyglot detection
- Macro/script detection and stripping

L3 Adaptive (Slow Path, 50-200ms):

- Full document rendering → OCR → compare against raw extraction
- Embedded image extraction → image security pipeline
- Hyperlink and external reference scanning
- Document structure anomaly detection

4.6 Tool Use Security Module

L2 Innate (Fast Path, <5ms):

- Function definition scanning: regex patterns for injection indicators in tool descriptions
- Tool call authorization: verify against capability manifest
- Schema validation of tool call parameters

L3 Adaptive (Slow Path, 10-50ms):

- Semantic analysis of tool descriptions for hidden instructions
- Behavioral baselining of tool call sequences
- Cross-tool chain analysis: detect multi-step attack patterns
- Tool output scanning before results feed back to model

4.7 Cross-Modal Correlation Engine (New L3 Module)

A dedicated analyzer that examines the relationship between all modalities present in a single request:

- Semantic consistency scoring: do all modalities tell a coherent story, or is there evidence of modality laundering?
- Multi-modal DCA signal fusion: aggregate PAMP, danger, and safe signals across all modalities
- Cross-modal escalation tracking: maintain per-session state across modalities for progressive attack detection

5. Implementation Priority Matrix

Priority	Module	Justification	Effort
P0	Image OCR → text pipeline	Highest frequency attack vector; reuses existing L2/L3 scanners	Low
P0	Tool use authorization	Agent security is the 2026 market; directly supports Stack vision	Medium
P1	Image steganalysis + sanitization	Addresses invisible injection attacks	Medium
P1	Document hidden text detection	PDF/doc injection is common in enterprise workflows	Low-Medium
P1	Audio transcription → text pipeline	Reuses existing scanners; critical for voice-enabled models	Low
P2	Image adversarial perturbation detection	Requires ML model training; lower current frequency	High
P2	Audio WaveGuard defense	Requires signal processing expertise	Medium-High
P2	Cross-modal correlation engine	Novel capability; competitive differentiator	High

Priority	Module	Justification	Effort
P3	Video frame analysis	Lower current deployment; depends on image module	Medium
P3	Voice liveness detection	Specialized use case (authentication)	Medium

6. Future-Proofing Considerations

6.1 Anticipating Attacker Evolution

The attack research trajectory is clear: attacks become more sophisticated along three axes simultaneously:

- From visible to invisible (typographic → steganographic → adversarial perturbation)
- From single-modal to cross-modal (text-only → image+text → coordinated multi-modal)
- From single-turn to multi-step (direct injection → progressive escalation → autonomous agent chains)

AEGIS's immune system architecture naturally handles this evolution. The DCA's multi-signal fusion is modality-agnostic — it doesn't care whether the PAMP signal came from a text regex match or an image steganalysis detection. The Danger Theory approach (detect anomaly, not specific attack) is inherently future-proof because it doesn't require enumerating every possible attack in every modality.

6.2 Embedding Model Strategy

The threat vault currently uses MiniLM-L6-v2 for text embeddings. Multimodal extension requires:

- CLIP or SigLIP for image embeddings (shared text-image embedding space enables cross-modal similarity search)
- Whisper encoder for audio embeddings
- A unified embedding approach (e.g., ImageBind from Meta, which maps six modalities into a single embedding space) as the long-term target

The critical insight: a shared embedding space means the threat vault can find similarities across modalities. A text-based injection pattern that gets embedded in the vault would also catch a semantically similar image-based injection, even if the vault entry was created from a text attack.

6.3 Emerging Modalities to Monitor

- 3D/spatial inputs (AR/VR applications, robotics)
- Sensor/telemetry data (IoT, autonomous vehicles)
- Code as input (code generation models processing arbitrary code)
- Brain-computer interfaces (speculative but emerging)

- Haptic/tactile inputs (robotics, accessibility)

6.4 Regulatory Alignment

The EU AI Act (August 2026 enforcement) does not distinguish between modalities in its security requirements. Article 15 (cybersecurity) applies equally to text, image, audio, and video processing. Building multimodal security before the enforcement date positions AEGIS ahead of regulatory requirements.

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8. Conclusion

The multimodal threat landscape is not a future concern — it is a present reality with published attack success rates exceeding 80% in some categories. AEGIS's immune system architecture is uniquely positioned to address these threats because the scanner module pattern, the DCA multi-signal fusion, and the Danger Theory approach are all modality-agnostic by design. What AEGIS needs is modality-specific scanner modules that normalize non-text inputs into forms the existing detection pipeline can process, combined with modality-specific deep analysis modules for threats that have no text equivalent.

The highest-priority extensions — OCR-based image scanning, tool use authorization, and document hidden text detection — reuse the existing L2/L3 text detection pipeline and can be implemented with moderate effort. More advanced capabilities — steganalysis, adversarial perturbation detection, cross-modal correlation — require dedicated ML models and represent the platform's next competitive frontier.

Building these capabilities before the market demands them — rather than after the first high-profile multimodal attack on a production system — is consistent with AEGIS's core philosophy: defense before frameworks, engineering before governance, and proactive hardening over reactive patching.

This document is proprietary to Odins LLC. It serves as the architectural research foundation for extending AEGIS multimodal security capabilities.